Scalability & Future Plan

Date	15 March 2026
Team ID	XXXXXX
Project Name	F1 Race Intelligence Dashboard
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Static dataset (6,464 race records) loaded once; no live data feed.	Dashboard does not show new races or seasons until the data is manually reloaded.	High
2	Data quality anomalies: season axis extends to 2040, negative points appear, average pit stop times are around 23 s.	Can mislead trend analysis and reduce trust in the results.	High
3	Global Driver Distribution shows 19 drivers for every country, and some charts are limited to 10 teams or the top 10 drivers.	Country field appears to reflect race location rather than driver nationality; limited coverage of the full grid.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Dashboard viewed by me and the evaluators from a single workbook.	Publish to Tableau Cloud/Server so many viewers can access it at once, with caching and scheduled extract refresh.
2	Data Storage	Single flat table (race_results) in a file-based source.	Move to a cloud database (PostgreSQL / BigQuery) with separate tables for drivers, teams, circuits and races.
3	Performance	Handles 6,464 rows smoothly with several charts on one page.	Use Hyper extracts, pre-aggregated views and indexed tables so the dashboard stays fast with decades of data.
4	Security	No access control; data is open to anyone with the file.	Add login (SSO), role-based access, read-only database credentials and encrypted connections.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Clean the data (fix season range, negative points, pit stop values) and connect a live race-data API for automatic updates.	Month 1 to 2	Accurate, always up-to-date dashboard.

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 3	Predictive analytics: race-finish and DNF-risk models using grid position, tyre choice and pit stop data.	Month 3 to 4	Moves from reporting to forecasting.
Phase 4	Public deployment with mobile layout, driver and team comparison tool, and alerts for new race results.	Month 5 to 6	Wider reach and better user experience.